

Lab 1

1. Identify Entities

The main entities identified in the Airport Database system are:

1. Airport
2. Airline
3. Flight
4. Passenger
5. Booking
6. Booking History
7. Boarding Pass
8. Baggage
9. Baggage Checking
10. Security Check

2. Determine Attributes

airport

Attribute	Data Type	Key
airport_id	INT	PK
airport_name	VARCHAR(100)	
country	VARCHAR(50)	
state	VARCHAR(50)	
city	VARCHAR(50)	
created_at	DATETIME	
updated_at	DATETIME	

airline

Attribute	Data type	Key
Airline_id	INT	PK
Airline_code	VARCHAR(10)	UNIQUE
Airline_name	VARCHAR(100)	
country	VARCHAR(50)	
Created_at	DATETIME	
Updated_at	DATETIME	

Flight

Attribute	Data type	Key
Flight_id	INT	PK
Departing_gate	VARCHAR(10)	
Arriving_gate	VARCHAR(10)	
Created_at	DATETIME	
Updated_at	DATEITME	
Airline_id	INT	FK
Departure_airport_id	INT	FK
Arrival_airport_id	INT	FK
Scheduled_departure_time	DATETIME	
Scheduled_arrival_time	DATETIME	
Actual_departure_time	DATETIME	
Actural_arrival_time	DATETIME	

Passenger

Attribute	Data type	Key
Passenger_id	INT	PK
First_name	VARCHAR(50)	
Last_name	VARCHAR(50)	
gender	VARCHAR(20)	
Date_of_birth	DATE	
Citizenship_country	VARCHAR(50)	
Residence_country	VARCHAR(50)	
Passport_numder	VARCHAR(30)	UNIQUE
Created_at	DATETIME	
Updated_at	DATETIME	

Booking

Attribute	Data type	Key
booking_id	INT	PK
Flight_id	INT	FK
Passenger_id	INT	FK
status	VARCHAR(30)	
Booking_platform	VARCHAR(50)	
Ticket_price	DECIMAL(10,2)	
Created_at	DATETIME	
Updated_at	DATETIME	

Boarding pass

Attribute	Data type	Key
Boarding_pass_id	INT	PK
Booking_id	INT	FK
seat	VARCHAR(10)	
Boarding_time	DATETIME	
Created_at	DATETIME	
Updated_at	DATETIME	

Baggage

Attribute	Data type	Key
Baggage_id	INT	PK
Booking_id	INT	FK
Weight_kg	DECIMAL(5,2)	
Created_at	DATETIME	
Updated_at	DATETIME	

Booking history

Attribute	Data type	Key
Booking_history_id	INT	PK
Booking_id	INT	FK
Old_flight_id	INT	FK
New_flight_id	INT	FK
Change_at	DATETIME	

Baggage checking

Attribute	Data type	Key
Baggage_checking_id	INT	PK
Booking_id	INT	FK
Passenger_id	INT	FK
Check_result	VARCHAR(50)	
Created_at	DATETIME	
Updated_at	DATETIME	

Security check

Attribute	Data type	Key
Security_check_id	INT	PK
Passenger_id	INT	FK
Check_result	VARCHAR(50)	
Created_at	DATETIME	

Updated_at	DATETIME	
------------	----------	--

3. Relationships and Cardinality

Relationship	Cardinality
Airline→flight	1:M
Airport→flight	1:M
Passenger→ booking	1:M
Flight→ booking	1:M
Booking→boarding pass	1:1
Booking→ baggage	1:M
Booking→ booking history	1:M
Booking → baggage checking	1:M
Passenger→baggage checking	1:M
Passenger→security check	1:M